

Name: _____

These questions assess your understanding of the material from Chapter(s) 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units. Correct numerical answers do not guarantee credit.

1. An electron gun has parallel plates separated by **1.8 cm** and gives electrons **70. keV** of energy. What force would the electric field between the plates have on a piece of plastic that has a charge of **0.28 μC** ?

ANSWER: _____

2. Suppose **4.5 mm** separates two **1.1-m²** plates in a parallel-plate capacitor. What charge is stored in this capacitor if it is connected to a **1.8 -V** battery?

ANSWER: _____

3. Dry air will support electric fields up to its dielectric strength, **$3.0 \times 10^6 \frac{\text{V}}{\text{m}}$** . Above that value, an electric field creates enough ionization in the air to make the air a conductor, which causes a discharge or spark to reduce the field. Calculate the maximum voltage between a charged region at the base of a thundercloud and the ground if they are separated by **1.3 km** of dry air. Model the cloud and ground as two parallel conducting plates.

ANSWER: _____

4. A generator coil is rotated 0.30 revolutions in 36. ms. The 140-turn circular coil has a 2.3-cm radius and is in a uniform 1.2-T magnetic field. What is the maximum emf that is induced?

ANSWER: _____

5. Calculate the sound intensity level in decibels for a sound wave traveling in air at 9.3°C with a pressure amplitude of 1.68 Pa. Recall that the density of air is $1.29 \frac{\text{kg}}{\text{m}^3}$ and the speed of sound in air at 0°C is $331 \frac{\text{m}}{\text{s}}$.

ANSWER: _____

6. A car headlight filament is made of tungsten ($\rho = 5.6 \times 10^{-8} \Omega \cdot \text{m}$) and has a resistance of 0.20 Ω . If the filament is a cylinder 1.5-cm long, what is its diameter?

ANSWER: _____

7. Suppose you pass light from a laser through two slits separated by 0.011 mm and find that the third bright line on a screen is formed at an angle of $39.^\circ$ relative to the incident beam. What is the wavelength of the light?

ANSWER: _____

8. Suppose you have a tube that is closed at one end. If you want its fundamental frequency to be $175. \text{ Hz}$ and the air temperature is 0.51°C , then what should the length of the tube be?

ANSWER: _____

9. A clear-glass light bulb is placed 0.43 m from a thin convex lens that has a focal length of 0.12 m . Calculate the magnification of the produced image.

ANSWER: _____

10. Three charges lie along the x-axis. The positive charge $q_1 = 20. \mu C$ is at $x_1 = 1.9 \text{ m}$ and the positive charge $q_2 = 14. \mu C$ is at the origin $x_2 = 0 \text{ m}$. Where must a negative charge q_3 be placed on the x-axis such that the resultant force on it is zero?

ANSWER: _____

11. Calculate the critical angle for light traveling in a material that is surrounded by air, given that the material has an index of refraction of 1.6.

ANSWER: _____

12. A medical imaging device produces ultrasound by oscillating with a period of $0.229 \mu s$. What is the frequency of this oscillation?

ANSWER: _____

13. A battery charger meant for a series connection of several nickel-cadmium batteries needs to have a $16. \text{ V}$ output to charge the batteries. It uses a step-down transformer with a $91. \text{ -loop}$ primary and a 130 -V input. If the charging current is $29. \text{ A}$, what is the input current?

ANSWER: _____

14. Suppose a space probe is moving away from Earth at a speed of $1.3 \times 10^8 \text{ m/s}$. It sends a radio-wave message back to Earth at a frequency of 0.74 GHz . At what frequency is the message received on Earth?

ANSWER: _____

15. A proton travels with speed $2.4 \times 10^6 \frac{\text{m}}{\text{s}}$ at an angle of $37. \square^\circ$ to a uniform magnetic field. If its path has a circular component of radius 0.072 m , calculate the magnitude of the magnetic field.

ANSWER: _____

16. How much energy is stored in a 1.5 -mH inductor when a 38. -A current flows through it?

ANSWER: _____

17. Ultrasound is transmitted from fat to muscle during a particular medical procedure. Calculate the percentage of the ultrasound's intensity that is reflected when the ultrasound travels from fat with a density of $868. \frac{\text{kg}}{\text{m}^3}$ to muscle tissue with a density of

$1.022 \times 10^3 \frac{\text{kg}}{\text{m}^3}$. The speed of ultrasound in the fat is $1.49 \times 10^3 \frac{\text{m}}{\text{s}}$ and the speed of ultrasound in the muscle tissue is $1.53 \times 10^3 \frac{\text{m}}{\text{s}}$.

ANSWER: _____

18. What is the resistance of an automobile headlight which carries a current of 1.1 A when 6.4 V is applied to it?

ANSWER: _____

19. Suppose that you take a magnifying glass out on a sunny day and you find that it concentrates sunlight to a small spot **13 . cm** away from the lens. What is the power of the lens?

ANSWER: _____

20. What is the maximum strength of the magnetic field in an electromagnetic wave that has a maximum electric-field strength of **390 V/m**?

ANSWER: _____

21. A portable x-ray unit has a step-up transformer, the 120 V -input of which is transformed to the $3.4 \times 10^5\text{-V}$ output needed by the x-ray tube. The primary has $26.$ loops and draws a current of 7.4 A when in use. Find the current output of the secondary.

ANSWER: _____

22. What is the length of a pendulum that has a period of 0.38 s ?

ANSWER: _____

23. Find the total capacitance for three capacitors connected in series, given that their individual capacitances are 3.0 nF , $12.\text{ nF}$, and $26.\text{ nF}$.

ANSWER: _____

24. A 12.8-V car battery supplies a starter motor drawing 750 A for 1.8 s. Calculate the total charge moved by the battery during this time.

ANSWER: _____

25. Suppose you have a tube that is closed at one end. Its length is 37. cm and the air temperature is 2.7°C. What is the frequency of its second overtone?

ANSWER: _____

26. Suppose that you rub a glass rod with silk, placing a 14. -nC positive charge on it. Calculate the initial force on the rod due to the Earth's magnetic field, if you throw the rod with a horizontal velocity of $2.6 \frac{\text{m}}{\text{s}}$ due west in a place where Earth's field is due north. Recall that the magnitude of Earth's magnetic field is 37. μT .

ANSWER: _____

27. While starting an engine, a truck battery sets in motion $1.3 \times 10^3 \text{ C}$ of charge in 1.1 s. Calculate the current involved in starting the engine.

ANSWER: _____

28. When James gets in his car, it lowers by 2.0 cm. If James's mass is 120 kg, then what is the force constant of his car's suspension system?

ANSWER: _____

29. Calculate the power of the eye when viewing an object 11. km away. Take the lens-to-retina distance to be 1.9 cm.

ANSWER: _____

30. What is the magnitude of the maximum force exerted on an aluminum rod with a $0.40\text{-}\mu\text{C}$ charge that you pass between the poles of a 0.54-T permanent magnet at a speed of $18.\frac{\text{m}}{\text{s}}$?

ANSWER: _____

31. Suppose the voltage supplied to a circuit is 5.3 V . There are three resistors wired in parallel: $R_1 = 3.6\ \Omega$, $R_2 = 2.0\ \Omega$, and $R_3 = 6.3\ \Omega$. Calculate the current leaving the battery.

ANSWER: _____

32. What current flows through a high-voltage transmission line to transmit $30.\text{ MW}$ of power at 400 kV ?

ANSWER: _____

33. A large water main is 0.98 m in diameter and the average water velocity is $2.4\frac{\text{m}}{\text{s}}$. Find the Hall voltage produced if the pipe runs perpendicular to Earth's $6.0 \times 10^{-5}\text{-T}$ field.

ANSWER: _____

34. Suppose a 0.23-H inductor is in series with a $0.76\text{-}\Omega$ resistor and a current of 1.3 A flows through them. Find the current that flows through the inductor and resistor 7.9 s after the battery is disconnected.

ANSWER: _____

35. You shine a laser through air at an incident angle of $39.^\circ$ into an unknown material and measure the angle of refraction to be $33.^\circ$. What is the unknown material's index of refraction?

ANSWER: _____

36. What power of spectacle lens is needed to correct the vision of a near-sighted person whose far point is $24. \text{ cm}$? Assume the spectacle lens is held 1.4 cm away from the eye by eyeglass frames.

ANSWER: _____

37. Calculate the motional emf induced along a $31. \text{ km}$ -long conducting object moving at an orbital speed of $9.6 \frac{\text{km}}{\text{s}}$ perpendicular to Earth's $32. \text{ }\mu\text{T}$ magnetic field.

ANSWER: _____

38. Solve for the current that passes through resistor R_4 given $V_1 = 7 \text{ V}$, $V_2 = 6 \text{ V}$, $R_1 = 2 \Omega$, $R_2 = 2 \Omega$, $R_3 = 3 \Omega$, $R_4 = 8 \Omega$.

ANSWER: _____

39. A 11.8-V car battery with internal resistance $0.013\,\Omega$ supplies a starter motor drawing 510 A for 1.2 s . Calculate the electrical energy delivered to the starter during this time.

ANSWER: _____

40. Calculate the minimum thickness of an oil slick on water that appears blue when illuminated by white light perpendicular to its surface. Take the blue wavelength to be 440 nm , $n_{\text{oil}} = 1.42$ and $n_{\text{water}} = 1.34$.

ANSWER: _____

41. Suppose the voltage supplied to a circuit is 5.2 V . There are three resistors wired in series: $R_1 = 8.8\,\Omega$, $R_2 = 4.7\,\Omega$, and $R_3 = 6.1\,\Omega$. Calculate the power output of the battery.

ANSWER: _____

42. A wire carries a 230 A current toward the south. If a large magnet were placed directly to the west of the wire and emits a 0.61 T magnetic field perpendicularly towards the wire, calculate the force exerted on the wire per length of wire.

ANSWER: _____

43. When a 1.7 -V battery runs a single 114. -W light, how many electrons pass through it each second?

ANSWER: _____

44. If the shock absorbers in a car go bad, then the car will oscillate at the smallest provocation, such as when going over bumps in the road and after stopping. Calculate the frequency of oscillations for a car with shoddy shock absorbers if the car's total mass is 884 . kg and the force constant of the suspension system is $1.57 \times 10^4 \frac{\text{N}}{\text{m}}$.

ANSWER: _____

45. Visible light of wavelength 430 nm falls on a single slit and produces its second diffraction minimum at an angle of $38.^\circ$ relative to the incident direction of the light. What is the width of the slit?

ANSWER: _____

46. Calculate the power of the eye when viewing an object $20.\text{ cm}$ away. Take the lens-to-retina distance to be 2.1 cm .

ANSWER: _____

47. What is the characteristic time constant for a $11.\text{ mH}$ inductor in series with a $0.70\text{ }\Omega$ resistor?

ANSWER: _____

48. Suppose a 180-turn coil lies in the plane of the page in a uniform magnetic field that is directed into the page. The coil originally has an area of 0.31 m^2 . It is then stretched for 0.35 s until it has no area. What is the magnitude of the induced emf if the uniform magnetic field has a strength of 0.60 T ?

ANSWER: _____

49. Calculate the self-inductance of a 0.080 m -long, 0.026 m -diameter solenoid that has 80. coils.

ANSWER: _____

50. Calculate the energy that is released upon the complete annihilation of a 1.3-g object.

ANSWER: _____

51. Calculate the strength of an electric field created by a 0.20-nC charged particle at a distance of 6.2 mm away.

ANSWER: _____

52. Calculate the final speed of a $2.6 \times 10^{-18}\text{-C}$ charged particle that accelerates from rest through a potential difference of 1.1 V . The mass of the charged particle is $2.5 \times 10^{-29}\text{ kg}$.

ANSWER: _____

53. A scientist at a natural-history museum is holding a specimen $24.\text{cm}$ from a thin convex magnifying glass of focal length $15.\text{cm}$. What magnification is produced?

ANSWER: _____

54. An electron moves at $2.4 \times 10^6 \frac{\text{m}}{\text{s}}$ perpendicular to a magnetic field. The field causes the electron to travel in a circular path of radius 13 . cm. What is the strength of the magnetic field?

ANSWER: _____

55. Solve for the current that passes through resistor R_1 given $V_1=5\text{ V}$, $V_2=6\text{ V}$, $V_3=6\text{ V}$, $R_1=1\ \Omega$, $R_2=2\ \Omega$, $R_3=4\ \Omega$, $R_4=6\ \Omega$, $R_5=3\ \Omega$, $R_6=2\ \Omega$, and $R_7=3\ \Omega$.

ANSWER: _____

56. The primary mirror of an orbiting space telescope has a diameter of 3.0 m. One of the key advantages of operating in space is that it avoids atmospheric distortion, allowing the telescope to approach its diffraction-limited resolution. Assuming the telescope observes visible light with a wavelength of 490 nm, what is the minimum angular separation between two point sources that can be just resolved according to the Rayleigh criterion? Report your answer in arcseconds. Note: 1 degree is 60 arcminutes and 1 arcminute is 60 arcseconds.

ANSWER: _____

57. Suppose that you would like to find the angle of refraction in a material. You shine a laser through air at an incident angle of 4.1° into the material which has an index of refraction of 1.3 . Calculate the angle of refraction in the material.

ANSWER: _____

58. A 2.4-m -long copper wire of diameter 0.74 mm is connected to a $13.\text{-V}$ battery at 4.6°C . Calculate the percent change in the power delivered by the wire when it is heated to $85.\text{ }^\circ\text{C}$. The resistivity of copper at 4.6°C is $1.7 \times 10^{-8} \Omega \cdot \text{m}$ and its linear thermal expansion coefficient is $3.9 \times 10^{-3} \frac{1}{^\circ\text{C}}$.

ANSWER: _____

59. A particular Van de Graaff generator has a metal sphere of diameter $13.\text{ cm}$ that produces a voltage of $457.\text{ kV}$ near its surface. What excess charge resides on the sphere?

ANSWER: _____

60. What is the electric potential 7.2 cm away from the center of a metal sphere that has a 0.14 nC static charge?

ANSWER: _____

61. Calculate the speed at which light travels in a particular material given that the index of refraction of the material is 1.3 .

ANSWER: _____

62. A 0.18-kg baseball, pitched at $41.\frac{\text{m}}{\text{s}}$ horizontally and perpendicularly to Earth's horizontal $39.\text{-}\mu\text{T}$ field, has a $32.\text{-nC}$ charge on it. What distance is the ball deflected from its path by the magnetic force after it travels 7.4 m horizontally? Do not consider the displacement due to the weight force. You may assume that the ball does not hit the ground during this motion.

ANSWER: _____

63. A circuit has two branches: one branch has capacitors 1 and 2 while the second branch has capacitors 3 and 4. Capacitors 1 and 2 are connected in series and capacitors 3 and 4 are connected in series. The two branches are connected in parallel. Draw this circuit and find the total capacitance given that $C_1 = 21. \text{ nF}$, $C_2 = 4.1 \text{ nF}$, $C_3 = 18. \text{ nF}$, and $C_4 = 12. \text{ nF}$. Note: you must draw the circuit correctly to receive credit.

ANSWER: _____

64. The headlights of a car have a wavelength of about 580 nm and are 0.70 -m apart. Assuming the eye's pupil diameter is 0.43 cm , what is the maximum distance at which the two headlights can be resolved? Use the diffraction limit (Rayleigh criterion) for your calculation.

ANSWER: _____

65. Suppose a 1.1-V custom battery can move $3.7 \times 10^3 \text{ C}$ of charge and a 12-V car battery can move $5.8 \times 10^4 \text{ C}$ of charge. What is the ratio of work done (or energy delivered) from the car battery compared to the custom battery?

ANSWER: _____

66. An incandescent lightbulb has a 0.86-cm -long filament and operates at 115 V to deliver 23 W . Calculate the diameter of the bulb's filament, assuming it to be a cylindrical tungsten wire, given that the resistivity of tungsten is $5.6 \times 10^{-8} \Omega \cdot \text{m}$.

ANSWER: _____

67. On its highest power setting, a 590-W microwave oven projects microwaves onto a 0.40-m -by- 0.40-m area. Calculate the peak electric-field strength in the electromagnetic waves emitted by the microwave oven.

ANSWER: _____

68. Suppose that a $1.05 \times 10^3\text{-kg}$ car has a suspension system with a force constant $5.52 \times 10^4 \frac{\text{N}}{\text{m}}$. If the car hits a bump and bounces with an amplitude of 5.0 cm , then what is its maximum vertical velocity assuming no damping occurs?

ANSWER: _____

69. A generator coil is rotated **1.1 revolutions** in **7.3 ms**. The **110-turn** circular coil has a **1.5-cm** radius and is in a uniform **0.69-T** magnetic field. What is the average emf that is induced?

ANSWER: _____

70. A clear-glass light bulb is placed **0.43 m** from a thin convex lens that has a focal length of **0.13 m**. Calculate the distance from the lens to the location of the produced image.

ANSWER: _____

71. What is the size of the image on the retina of a **0.66 mm**-diameter strand of hair that is held at arm's length (**53. cm**) away? Take the lens-to-retina distance to be **1.6 cm**.

ANSWER: _____

72. Two identical charged spheres, each with a mass of 5.9 g , hang in equilibrium. They are suspended by a string of length 5.2 cm and are separated by a horizontal distance of $23.\text{ mm}$. Calculate the magnitude of the charge on each sphere.

ANSWER: _____

73. Suppose a cosmic ray colliding with a nucleus in Earth's upper atmosphere produces a muon that has a velocity equal to $1.7 \times 10^8\text{ m/s}$. The muon then travels at constant velocity and lives for a duration of $1.1\text{ }\mu\text{s}$ in the muon's frame of reference. How long does the muon live as measured by an Earth-bound observer?

ANSWER: _____

74. Calculate the radius of curvature of the path of a proton having a velocity of $1.6 \times 10^7 \frac{\text{m}}{\text{s}}$ perpendicular to a magnetic field of strength $B = 0.34 \text{ T}$.

ANSWER: _____

75. What angle would the axis of a polarizing filter need to make with the direction of polarized light of intensity 1.0 kW/m^2 to reduce the intensity to 0.10 kW/m^2 ?

ANSWER: _____

76. Calculate the magnitude of the force exerted on a 1.3 -nC charged particle placed in a $2.3 \times 10^4 \frac{\text{N}}{\text{C}}$ electric field.

ANSWER: _____

77. A circuit has two branches connected in parallel: branch A consists of resistors $R_1 = 3.9 \, \Omega$ and $R_2 = 4.4 \, \Omega$ in series; branch B consists of resistors $R_3 = 5.2 \, \Omega$ and $R_4 = 6.3 \, \Omega$ in parallel. Draw this circuit and find the total equivalent resistance. Note: you must draw the circuit correctly to receive credit.

ANSWER: _____

78. What power of spectacle lens is needed to correct the vision of a far-sighted person whose near point is $0.66 \, \text{m}$, so that they may see an object clearly that is $10. \, \text{cm}$ away? Assume the spectacle lens is held $1.8 \, \text{cm}$ away from the eye by eyeglass frames.

ANSWER: _____

79. Suppose an astronaut is traveling at $1.5 \times 10^8 \, \text{m/s}$ from Earth to Alpha Centauri, which is a star system $2.1 \, \text{ly}$ away as measured by an Earth-bound observer. How far apart are Earth and Alpha Centauri as measured by the astronaut? Note: ly is lightyear, the distance light travels in one year.

ANSWER: _____

80. Suppose a train sounding its 654.-Hz horn is moving at $55.4 \frac{\text{m}}{\text{s}}$. What frequency does a person standing by the tracks hear after the train passes? Given this particular day's temperature, the speed of sound is $332. \frac{\text{m}}{\text{s}}$.

ANSWER: _____

81. Find the current that must run through a long, straight wire so that it produces a 63.- μT magnetic field at a distance of 3.8 cm from the wire.

ANSWER: _____

82. Suppose a 0.286-kg object is connected to a horizontal spring. There is friction between the object and the surface, and the coefficient of kinetic friction μ_k is 0.149. What total distance does the object travel if it is released from rest at a distance of 11. cm from its equilibrium position? The force constant of the spring is $54.6 \frac{\text{N}}{\text{m}}$.

ANSWER: _____

83. A Hall-effect flow probe is placed on an artery, applying a 0.25 T magnetic field across it. What is the Hall emf, given the vessel's inside diameter is 3.1 mm and the average velocity of the blood is $21. \frac{\text{cm}}{\text{s}}$?

ANSWER: _____

84. Calculate the magnitude of the electrostatic force between an electron and a proton separated by 7.04 pm .

ANSWER: _____

85. The hot and neutral wires supplying DC power to a light-rail commuter train carry 860 A and are separated by $59. \text{ cm}$. What is the magnitude of the force between 100 m of these wires?

ANSWER: _____

86. How much energy is stored in the spring of a toy gun that has a force constant of $52.0 \frac{\text{N}}{\text{m}}$ and is compressed $14. \text{cm}$?

ANSWER: _____

87. Calculate the magnitude of an electric field measured at the origin due to the presence of two point charges q_1, q_2 . q_1 is located 0.15 cm south of the origin and q_2 is located 0.51 cm east of the origin. $q_1 = +8.4 \text{ nC}$ and $q_2 = -0.22 \text{ nC}$.

ANSWER: _____

88. A $36. \text{-m}$ -long copper wire must carry $29. \text{ A}$ without dissipating more than 130 W as heat. Calculate the wire's minimum diameter given that copper's resistivity is $1.7 \times 10^{-8} \Omega \cdot \text{m}$.

ANSWER: _____

89. Find the current that must run through a current-carrying circular loop so that it produces a $120\text{-}\mu\text{T}$ magnetic field at its center, given that the radius of the circular loop is 11 cm .

ANSWER: _____

90. A 12.5-V car battery supplies a starter motor drawing 290 A for 2.4 s . Calculate the electrical energy delivered by the battery during this time.

ANSWER: _____

91. You shine a beam of white light through a diffraction grating with 6.2×10^3 lines per centimeter to a screen 2.4 m away. Find the angle for first-order diffraction of 680 nm light.

ANSWER: _____

92. A sonar echo returns to a submarine 1.09 s after being emitted from the submarine. What is the distance from the submarine to the object, given that the speed of sound in this water is $1.51 \frac{\text{km}}{\text{s}}$?

ANSWER: _____

93. Suppose a galaxy is moving away from Earth at a speed of $1.4 \times 10^8 \text{ m/s}$. It emits radio waves with a wavelength of 0.42 m. What wavelength would we detect on Earth?

ANSWER: _____

94. A heart defibrillator delivers $4.5 \times 10^3 \text{ J}$ of energy by discharging a capacitor initially at $6.8 \times 10^4 \text{ V}$. What is its capacitance?

ANSWER: _____

95. Suppose a tiny drop of oil has a mass of 3.5 fg and is given a positive charge of $9.0 \times 10^{-17} \text{ C}$. Calculate the electric force on the drop if there is an upward electric field of strength $1.5 \times 10^4 \frac{\text{N}}{\text{C}}$ due to static electricity.

ANSWER: _____

96. Calculate the power dissipated by a $3.4\text{-}\Omega$ car headlight when it has a voltage applied to it of $14. \text{ V}$.

ANSWER: _____

97. Find the maximum torque on a 140-turn square loop of wire that is 6.5 cm on each side that carries 9.7 A of current in a 3.3-T field.

ANSWER: _____

98. What is the value of the peak voltage for 180-V AC power?

ANSWER: _____